

Exercise Sheet 1: Setup and Business-Cycle Statistics

Brigitte Hochmuth
Department of Economics
University of Copenhagen

This first sheet is practical. The goal is to get your setup working and produce a few simple descriptive statistics without being overwhelmed. The **main workflow uses the FRED API**. The fallback files are only there to keep the exercise moving if the API setup fails.

Use the ProgEcon material for help:

- <https://sites.google.com/view/numeconcph-progecon/home>
- especially the guides on installation, VS Code, APIs, and Excel/data work

Notebook files for this sheet:

- week01_task_bcstats.ipynb

0. Setup in VS Code

1. Open the task notebook in VS Code.
2. Make sure you can run Python cells.
3. Run the import cell.
4. If something fails, use the ProgEcon guides and fix your setup.

1. Download Data from FRED

Download at least two quarterly U.S. macroeconomic series from FRED. One should be real GDP. The other can for example be consumption or investment.

The ProgEcon material uses the `fredapi` package, and so do we here. You should create a FRED API key and use it in your notebook.

Suggested tasks:

1. download the series
2. put them in one `DataFrame`
3. clean missing values
4. make one simple line plot
5. save your downloaded data if you want
6. as a fallback, load the pre-downloaded FRED files in Python:
 - `week01_fred_gdp.xlsx`
 - `week01_fred_consumption.xlsx`
 - `week01_fred_investment.xlsx`

using `pd.read_excel()`

2. Summary Statistics

Create a small table with summary statistics for your main series:

- mean
- standard deviation
- minimum
- maximum

3. HP Filter

Choose one quarterly series and:

1. take logs
2. apply the HP filter with $\lambda = 1600$
3. store the trend and cycle

What does the HP filter do?

Let x_t denote the log of a macroeconomic time series, for example log real GDP. The HP filter decomposes the series as

$$x_t = \tau_t + c_t,$$

where τ_t is a smooth trend component and c_t is the cyclical component.

The trend is chosen by solving

$$\min_{\{\tau_t\}_{t=1}^T} \sum_{t=1}^T (x_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2.$$

The first term makes the trend fit the data, while the second term penalizes variation in the growth rate of the trend. For quarterly data, we typically use $\lambda = 1600$.

We use the HP filter because business-cycle statistics should be computed from c_t , not from the raw series x_t . Otherwise, long-run growth may dominate the comovement and volatility measures.

4. Plot Trend and Cycle

Produce one figure with:

- the log series and its trend, or
- the cyclical component

Keep the figure clean: title, axis labels, and legend if needed.

5. Business-Cycle Statistics

Compute a few simple business-cycle statistics. At minimum include:

- volatility of the cyclical component
- correlation between the cyclical component of GDP and the cyclical component of one other series

Then add one small extra subtask:

- restrict the sample to observations up to and including 2019Q4
- recompute the business-cycle statistics for this pre-COVID sample
- compare the full-sample and pre-2020 results

Short Questions

Answer briefly:

1. Which series is more volatile?
2. Is the correlation positive or negative?
3. What difference do you see between the raw series and the HP-filtered cycle?
4. Why is it useful to remove the trend before computing business-cycle statistics?
5. Does the sample ending in 2019 change your conclusion about whether consumption is more or less volatile than GDP?
6. Why might the COVID lockdown period affect consumption and therefore the business-cycle statistics?